SOP Number and version: SOP-TL-BSL2.v01

SOP Title: **Biosafety Level 2 Laboratory**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
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Effective Date:	**09/01/2025**
Review Date	09/01/2025

Approved by PI (Frank J. Hernandez) - 2026-01-28T12:39:53.937Z

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1. Overview and background

Biosafety is the application of safety precautions that reduce a laboratorian's risk of exposure to a potentially infectious microbe and limit contamination of the work environment and, ultimately, the community. The biosafety levels range from BSL-1 to BSL-4 **(Figure 1).** Each level has specific controls for containment of microbes and biological agents and each biosafety level builds on the controls of the level before it. The primary risks that determine levels of containment are infectivity, severity of disease, transmissibility, and the nature of the work conducted. The origin of the microbe, or the agent in question, and the route of exposure are also important.

In a Biosafety Level 2 (BSL-2) laboratory, the microbes present pose moderate hazards to laboratorians and the surrounding environment. These microbes are generally indigenous and can cause diseases of varying severity in humans. A common example of a microbe studied in a BSL-2 laboratory is *Staphylococcus aureus*.

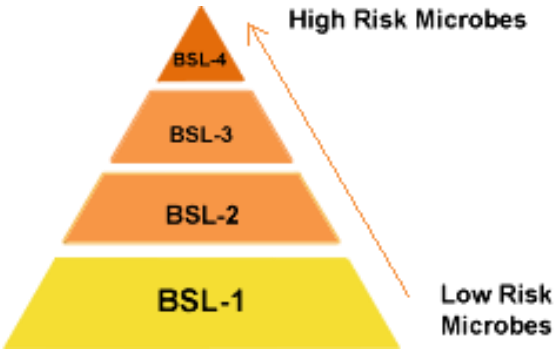

Figure 1. Biosafety Levels

In addition to BSL-1 considerations, BSL-2 laboratories have the following containment requirements:

- Access to the laboratory is restricted when work is being conducted.
- Appropriate personal protective equipment should be used, including lab coats and gloves. Eye protection and face shields can also be worn, as needed.
- All procedures that can cause infection from aerosols or splashes are performed within a biological safety cabinet.
- An autoclave or an alternative method of decontamination is available for proper disposal.
- The laboratory has self-closing doors.
- A sink and eyewash are readily available.

2. Scope

This SOP outlines the safe and effective use of the Biosafety Level 2 (BSL-2) laboratory at Theralia R&D Laboratories, located at Linköping University (Campus Valla, B-huset, Rum 2E:686). It applies to all authorized personnel involved in activities that require handling indigenous microbes posing moderate hazards to humans and the environment. This document establishes guidelines for safety practices, equipment usage, waste management, and adherence to biosafety protocols to minimize risks associated with laboratory work.

3. Procedure

Access to the laboratory is limited to trained staff with permission of the CSO, Frank Hernandez. All staff working in this laboratory must be advised of potential hazards and sign the appendix of this SOP indicating this awareness and complete training.

- All staff must wear gloves.
- Contaminated gloves must be changed immediately. Under no circumstances will gloves be reused.
- Gloves can never be worn when touching doorknobs or other items that will leave this room.
- All staff must wash their hands after handling infectious materials, after removing gloves, and before leaving the laboratory as described in **SOP-TL-STER**
- Laboratory staff must never eat, smoke, handle contact lenses, or apply cosmetics in the laboratory.
- Food for human consumption should be stored outside the laboratory work area in refrigerators designated for that purpose only.
- Only mechanical pipetting devices should be used in the laboratory.
- All sharps must be placed in sharps containers as described in **SOP-TL-STER**. Sharps containers must be turned in when ¾ full.
- Forceps, a brush and pan or other mechanical means should be used to pick up broken glass. Broken glass may not be picked up by hand. Broken glass must be placed in a sharps container.

- All non-liquid contaminated cultures, stocks, and other biologically contaminated waste must be placed in biohazard waste containers for disposal as described in **SOP-TL-STER.**
- Inactivation of liquid biological waste can be achieved by autoclaving (121 ^{o}C for 15-20 minutes or 115 ^{o}C for 30 minutes) or by treating the solutions with an appropriate chemical decontaminant (e.g. 70% ethanol, <5% sodium hypochlorite) prior to being poured into drain as described in **SOP-TL-STER.**
- All procedures must be performed to minimize the creation of splashes or aerosols.
- All work surfaces and equipment must be decontaminated at the completion of work, at the end of the day, or after any spill or splash of viable material as described in **SOP-TL-STER.**
- Persons who are at increased risk of infection, or for whom infection may have serious consequences, must not be allowed to enter the laboratory when work with infectious agents is in progress.
- The Biosafety Cabinet must be used for all procedures with potential for creating infectious aerosols or splashes, or whenever handling high concentrations of infectious materials.
- Face protection (goggles, mask, face shield, or other splatter guard) must be used for all procedures when such procedures could produce splashes or sprays of infectious or other hazardous materials.
- Lab coats must be removed and left in the laboratory before leaving the room.
- Lab coats must be autoclaved prior to being washed and never be taken home.
- Laboratory doors must be locked when the laboratory is not occupied.
- Spaces between benches, cabinets, and equipment must be kept accessible for cleaning.
- Lab benches, shelving and drawers must be kept clean, organized and uncluttered.
- Eyewash must be flushed at least monthly and kept clear of obstructions.
- All personnel must know the location of the nearest safety shower.

4. Purpose/Objective

The objective of this SOP is to guide Theralia staff on how to use BSL-2 laboratory in Theralia R&D Laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- R&D – Research and Development
- CTO – Chief Technology Officer
- CSO – Chief Scientific Officer
- COO – Chief Operations Officer
- BSL – Biosafety Level

6. Responsibilities

This SOP was prepared by CTO, Baris Ata Borsa, revised by the CSO, Frank Hernandez and approved by the COO, Luiza I. Hernandez

7. Equipment/Material

- Autoclave
- Autoclave gloves
- Double-layer biohazard waste bags
- Sharps container
- Hand disinfectants
- Surface disinfectants
- Gloves
- Paper towel
- Water
- Soap

8. Caution/Warning

- Any accidents must be immediately reported to the Principal Investigator.

9. References

- https://www.cdc.gov/training/quicklearns/biosafety/
- World Health Organization. Infection prevention and control of epidemic- and pandemic-prone acute respiratory infections in health care. (2014). ISBN 978 92 4 150713 4.
- https://liu.se/en/lab-safety/waste/laboratory-waste

10. Version Control

Original version of the **SOP-TL-BSL2.v01**. Check **Table 1**.

11.Attachments/Appendices

Appendix I:

- Autoclave manual
- Biosafety cabinet manual
- User signature sheet